Thermally-Weighted Interband JDOS at the X and L Critical Points of Gold

Reduction of the 3D Emission Integral to One Dimension

Following Guerrisi, Rosei & Winsemius, Phys. Rev. B ***12****, 557 (1975)*

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1 Starting Point

The direct interband emission rate at photon energy $\hbar\omega$ is given by Fermi's golden rule,

$$\Gamma_e^{\mathrm{cv}}(\hbar\omega) \;\propto\; \int_{\mathrm{BZ}} \mathrm{d}^3k\; f\big(\mathcal{E}_c(\mathbf{k})\big)\,\big[1 - f\big(\mathcal{E}_v(\mathbf{k})\big)\big]\;\delta\big(\mathcal{E}_c(\mathbf{k}) - \mathcal{E}_v(\mathbf{k}) - \hbar\omega\big), \tag{1}$$

where $f(\mathcal{E}) = [\,1+e^{\mathcal{E}/k_BT}\,]^{-1}$ is the Fermi–Dirac distribution with energies measured from $E_F = 0$. We assume direct transitions ($\mathbf{k}_1 \approx \mathbf{k}_2$) so both bands are evaluated at the same $\mathbf{k}$.

1.1 Band Dispersions

Near each critical point the bands are expanded to second order with cylindrical symmetry ($k_\perp$ transverse to the symmetry axis, $k_\parallel$ along it):

$$\boxed{\mathcal{E}_v(\mathbf{k}) = -\mathcal{E}_{0v} - A_v k_\perp^2 - B_v k_\parallel^2, \qquad \mathcal{E}_c(\mathbf{k}) = \mathcal{E}_{0c} + A_c k_\perp^2 - B_c k_\parallel^2,} \tag{2}$$

with all coefficients A_c, B_c, A_v, $B_v > 0$. The conduction band is a *saddle point* at both X and L: it curves upward along $k_\perp$ and downward along $k_\parallel$.

Table 1: Band parameters at the two critical points. Curvature coefficients $A = \hbar^2/(2m_\perp^*)$, $B = \hbar^2/(2m_\parallel^*)$ with $C \equiv \hbar^2/(2m_e) = 3.81$ eV$\cdot$Å^2.

Point	N	$\mathcal{E}_{0c}$	$\mathcal{E}_g$	Topology	Axes
X (Δ)	6	0 (ref.)	$\sim$1.94 eV	Saddle (M_1)	$k_\perp$: BZ face; $k_\parallel$: along Δ
L (Λ)	8	$\mathcal{E}_{0c}^L \neq 0$	$\sim$2.45 eV	Saddle (M_1)	$k_\perp$: hex face; $k_\parallel$: along Λ

The derivation below is **carried out once in general form**. All expressions apply identically to both critical points—only the numerical values of $(A_c, B_c, A_v, B_v, \mathcal{E}_{0c}, \mathcal{E}_g, N)$ differ.

2 Change of Variables I: Cylindrical Symmetry

Exploiting the azimuthal symmetry around the symmetry axis,

$$\mathrm{d}^3k = \mathrm{d}k_x\,\mathrm{d}k_y\,\mathrm{d}k_z \;\longrightarrow\; k_\perp\,\mathrm{d}k_\perp\,\mathrm{d}k_\parallel\,\mathrm{d}\phi \;\longrightarrow\; 2\pi\,k_\perp\,\mathrm{d}k_\perp\,\mathrm{d}k_\parallel. \tag{3}$$

3 Change of Variables II: Linearization of Momenta

Define $u \equiv k_\perp^2 \geq 0$ and $v \equiv k_\parallel^2 \geq 0$:

$$\mathrm{d}u = 2k_\perp\,\mathrm{d}k_\perp, \qquad \mathrm{d}v = 2|k_\parallel|\,\mathrm{d}k_\parallel \;\;\Rightarrow\;\; \mathrm{d}k_\parallel = \frac{\mathrm{d}v}{2\sqrt{v}}. \tag{4}$$

The factor of 2 from the $\pm k_\parallel$ symmetry cancels the 1/2, giving

$$2\pi\, k_\perp\, \mathrm{d}k_\perp\, \mathrm{d}k_\parallel \;\longrightarrow\; \pi\, \frac{\mathrm{d}u\, \mathrm{d}v}{\sqrt{v}}, \qquad u \ge 0, \;\; v \ge 0. \tag{5}$$

In these variables the dispersions are *linear*:

$$\mathcal{E}_c(u,v) = \mathcal{E}_{0c} + A_c u - B_c v, \qquad \mathcal{E}_v(u,v) = -\mathcal{E}_{0v} - A_v u - B_v v. \tag{6}$$

4 Change of Variables III: Energy Space

We introduce two physically motivated variables:

$$\mathcal{E} \;\equiv\; \mathcal{E}_c(u,v), \qquad \Delta \;\equiv\; \mathcal{E}_c(u,v) - \mathcal{E}_v(u,v), \tag{7}$$

so that

$$\mathcal{E} - \mathcal{E}_{0c} = A_c\, u - B_c\, v, \tag{8}$$
$$\Delta - \mathcal{E}_g = \bar{A}\, u + \bar{B}\, v, \tag{9}$$

where

$$\boxed{\mathcal{E}_g \equiv \mathcal{E}_{0c} + \mathcal{E}_{0v}, \qquad \bar{A} \equiv A_c + A_v, \qquad \bar{B} \equiv B_v - B_c.} \tag{10}$$

4.1 Matrix Form and Jacobian

Equations (8)–(9) in matrix form:

$$\begin{pmatrix} \mathcal{E} - \mathcal{E}_{0c} \\ \Delta - \mathcal{E}_g \end{pmatrix} = \underbrace{\begin{pmatrix} A_c & -B_c \\ \bar{A} & \bar{B} \end{pmatrix}}_{M} \begin{pmatrix} u \\ v \end{pmatrix}. \tag{11}$$

The Jacobian determinant is

$$D \;\equiv\; \det M = A_c \bar{B} + B_c \bar{A} = A_c(B_v - B_c) + B_c(A_c + A_v) = \boxed{A_c B_v + A_v B_c \;>\; 0.} \tag{12}$$

Since all curvature coefficients are positive, $D > 0$ always. No sign ambiguity arises.

4.2 Inverse Transformation

$$\begin{pmatrix} u \\ v \end{pmatrix} = \frac{1}{D} \begin{pmatrix} \bar{B} & B_c \\ -\bar{A} & A_c \end{pmatrix} \begin{pmatrix} \mathcal{E} - \mathcal{E}_{0c} \\ \Delta - \mathcal{E}_g \end{pmatrix}, \tag{13}$$

giving explicitly

$$u(\mathcal{E}, \Delta) = \frac{1}{D}\Big[\bar{B}\,(\mathcal{E} - \mathcal{E}_{0c}) + B_c\,(\Delta - \mathcal{E}_g)\Big], \tag{14}$$

$$v(\mathcal{E}, \Delta) = \frac{1}{D}\Big[-\bar{A}\,(\mathcal{E} - \mathcal{E}_{0c}) + A_c\,(\Delta - \mathcal{E}_g)\Big]. \tag{15}$$

5 Integration Limits

The physical constraints $u \geq 0$ and $v \geq 0$ with $D > 0$ yield:

From $v \geq 0$ (eq. 15):

$$\bar{A}\,(\mathcal{E} - \mathcal{E}_{0c}) \;\leq\; A_c\,(\Delta - \mathcal{E}_g) \quad\Longrightarrow\quad \boxed{\mathcal{E} \;\leq\; \mathcal{E}_{0c} + \frac{A_c}{\bar{A}}\,(\Delta - \mathcal{E}_g) \;\equiv\; \mathcal{E}_{\max}(\Delta).} \tag{16}$$

From $u \geq 0$ (eq. 14):

$$\bar{B}\,(\mathcal{E} - \mathcal{E}_{0c}) \;\geq\; -B_c\,(\Delta - \mathcal{E}_g) \quad\Longrightarrow\quad \boxed{\mathcal{E} \;\geq\; \mathcal{E}_{0c} - \frac{B_c}{\bar{B}}\,(\Delta - \mathcal{E}_g) \;\equiv\; \mathcal{E}_{\min}(\Delta).} \tag{17}$$

Physical interpretation: $\mathcal{E}_{\max}$ corresponds to $v = 0$ ($k_\parallel = 0$, the transverse circle); $\mathcal{E}_{\min}$ corresponds to $u = 0$ ($k_\perp = 0$, the symmetry axis). The integration domain $\mathcal{E}_{\min} \leq \mathcal{E} \leq \mathcal{E}_{\max}$ is non-empty for $\Delta > \mathcal{E}_g$, i.e., for photon energies above the gap threshold.

6 Transformed Measure

From eq. (15), v can be expressed in terms of $\mathcal{E}_{\max}$:

$$v(\mathcal{E}, \Delta) = \frac{\bar{A}}{D}\,\big(\mathcal{E}_{\max}(\Delta) - \mathcal{E}\big), \tag{18}$$

which is manifestly non-negative for $\mathcal{E} \leq \mathcal{E}_{\max}$. Substituting into the measure (5) and including the $1/D$ Jacobian factor:

$$\pi\,\frac{\mathrm{d}u\,\mathrm{d}v}{\sqrt{v}} \;\longrightarrow\; \frac{\pi}{\sqrt{\bar{A}\,D}}\,\frac{\mathrm{d}\Delta\;\mathrm{d}\mathcal{E}}{\sqrt{\mathcal{E}_{\max}(\Delta) - \mathcal{E}}}. \tag{19}$$

The integrable singularity $(\mathcal{E}_{\max} - \mathcal{E})^{-1/2}$ occurs at the **upper limit**, where $v = 0$ (the transverse circle in k-space).

7 Resolving the Delta Function

Substituting eq. (19) into eq. (1) and using $\mathcal{E}_v = \mathcal{E} - \Delta$:

$$\Gamma_e^{\mathrm{cv}}(\hbar\omega) \propto \frac{\pi}{\sqrt{\bar{A}\,D}} \iint f(\mathcal{E})\,\big[1 - f(\mathcal{E} - \Delta)\big]\;\delta(\Delta - \hbar\omega)\;\frac{\mathrm{d}\Delta\;\mathrm{d}\mathcal{E}}{\sqrt{\mathcal{E}_{\max}(\Delta) - \mathcal{E}}}. \tag{20}$$

The δ-function sets $\Delta = \hbar\omega$ and eliminates one integration, yielding the central result:

$$\boxed{\Gamma_e^{\mathrm{cv}}(\hbar\omega) = \frac{\pi^2\,|\mu|^2}{\hbar\,\sqrt{\bar{A}\,D}} \int_{\mathcal{E}_{\min}(\hbar\omega)}^{\mathcal{E}_{\max}(\hbar\omega)} \frac{f(\mathcal{E})\,\big[1 - f(\mathcal{E} - \hbar\omega)\big]}{\sqrt{\mathcal{E}_{\max}(\hbar\omega) - \mathcal{E}}}\;\mathrm{d}\mathcal{E}} \tag{21}$$

with

$$\mathcal{E}_{\max}(\hbar\omega) = \mathcal{E}_{0c} + \frac{A_c}{\bar{A}}\,(\hbar\omega - \mathcal{E}_g), \qquad \mathcal{E}_{\min}(\hbar\omega) = \mathcal{E}_{0c} - \frac{B_c}{\bar{B}}\,(\hbar\omega - \mathcal{E}_g). \tag{22}$$

For absorption (ε_2), the Fermi factor becomes $f(\mathcal{E} - \hbar\omega)\,[1 - f(\mathcal{E})]$; at equilibrium the two are related by detailed balance $\Gamma_{\mathrm{em}}/\Gamma_{\mathrm{abs}} = e^{-\beta\hbar\omega}$.

8 Numerical Regularization

The singularity at $\mathcal{E} = \mathcal{E}_{\max}$ is integrable but inconvenient numerically. The substitution

$$t = \sqrt{\mathcal{E}_{\max} - \mathcal{E}}, \qquad \mathrm{d}\mathcal{E} = -2t\,\mathrm{d}t, \qquad \mathcal{E} = \mathcal{E}_{\max} - t^2, \tag{23}$$

transforms (21) into a smooth integral:

$$\Gamma_e^{\mathrm{cv}}(\hbar\omega) = \frac{2\pi^2|\mu|^2}{\hbar\sqrt{\bar{A}\,D}} \int_0^{\sqrt{\mathcal{E}_{\max}-\mathcal{E}_{\min}}} f\big(\mathcal{E}_{\max} - t^2\big)\ \big[1 - f\big(\mathcal{E}_{\max} - t^2 - \hbar\omega\big)\big]\ \mathrm{d}t, \tag{24}$$

which is evaluated by Simpson quadrature on a uniform t-grid.

9 Verification: Zero-Temperature JDOS

At $T = 0$ the Fermi functions become step functions, and for $\hbar\omega$ just above threshold:

$$J(\hbar\omega) \propto \int_{\mathcal{E}_{\min}}^{\mathcal{E}_{\max}} \frac{\mathrm{d}\mathcal{E}}{\sqrt{\mathcal{E}_{\max} - \mathcal{E}}} = 2\sqrt{\mathcal{E}_{\max} - \mathcal{E}_{\min}} \ \propto\ \sqrt{\hbar\omega - \mathcal{E}_g}, \tag{25}$$

recovering the M_0-type onset ($\sqrt{\cdot}$ threshold). Note that while each individual band is a saddle (M_1), the *energy-difference* surface $\Delta(u,v) = \mathcal{E}_g + \bar{A}u + \bar{B}v$ is a minimum in (u,v) space (both coefficients positive for $B_v > B_c$), giving M_0 onset for the JDOS.

10 Application to Gold: X vs. L Parameters

Since both critical points share the same saddle topology and identical integral structure (21), the full interband ε_2 model is simply a weighted sum:

$$\varepsilon_2(\omega) = \frac{S}{\omega^2}\Big[\big|P_X/P_L\big|^2 N_X\, J_X(\omega,T)\ +\ N_L\, J_L(\omega,T)\Big] + \frac{A_D}{\omega^3}, \tag{26}$$

where $J_{X,L}$ are evaluated using (24) with the respective parameters below.

The **only** differences between the X and L contributions are:

1. Different effective masses ($m^*_{c\perp}$, $m^*_{c\|}$, $m^*_{v\perp}$, $m^*_{v\|}$).
2. Different gap energies $\mathcal{E}_g$.
3. $\mathcal{E}_{0c}^L \neq 0$ (whereas $\mathcal{E}_{0c}^X \equiv 0$ by convention).
4. Different BZ multiplicities ($N_X = 6$, $N_L = 8$).

Summary of Transformations

$$\underbrace{d^3k}_{\text{3D BZ}} \xrightarrow{\text{cyl.}} \underbrace{2\pi k_\perp\, dk_\perp\, dk_\|}_{\text{2D}} \xrightarrow{u,v} \underbrace{\pi\,\frac{du\,dv}{\sqrt{v}}}_{\text{linear}} \xrightarrow{\mathcal{E},\Delta} \underbrace{\frac{\pi}{\sqrt{\bar{A}D}}\ \frac{d\Delta\,d\mathcal{E}}{\sqrt{\mathcal{E}_{\max} - \mathcal{E}}}}_{\text{energy space}} \xrightarrow{\delta} \underbrace{\text{1D}}_{(21)}$$

References: M. Guerrisi, R. Rosei, and P. Winsemius, *Phys. Rev. B* **12**, 557 (1975). N. E. Christensen and B. O. Seraphin, *Phys. Rev. B* **4**, 3321 (1971).

Table 2: Derived quantities for the two critical points. All curvature coefficients $A_\alpha = C/m^*_{\alpha\perp}$, $B_\alpha = C/m^*_{\alpha\parallel}$.

Quantity	**X point**	**L point**
Multiplicity N	6	8
$\mathcal{E}_{0c}$	0 (reference)	fitted ($\neq 0$)
$\mathcal{E}_g$	$\sim$1.94 eV	$\sim$2.45 eV
$\bar{A} = A_c + A_v$	$A_c^X + A_v^X$	$A_c^L + A_v^L$
$\bar{B} = B_v - B_c$	$B_v^X - B_c^X$	$B_v^L - B_c^L$
$D = A_c B_v + A_v B_c$	$A_c^X B_v^X + A_v^X B_c^X$	$A_c^L B_v^L + A_v^L B_c^L$
Singularity	$(\mathcal{E}_{\max} - \mathcal{E})^{-1/2}$	$(\mathcal{E}_{\max} - \mathcal{E})^{-1/2}$
$\mathcal{E}_{\max}(\hbar\omega)$	$\mathcal{E}_{0c} + \dfrac{A_c}{A_c + A_v}(\hbar\omega - \mathcal{E}_g)$	
$\mathcal{E}_{\min}(\hbar\omega)$	$\mathcal{E}_{0c} - \dfrac{B_c}{B_v - B_c}(\hbar\omega - \mathcal{E}_g)$	